

Nesting of Loops:-

- Loop inside the another loop is called nesting of loops

```
for i in range(10):
    for j in range(5):
        print(i*j)
```

`print(\n)` → New line

✓

```
for i in range(1,5):
    for j in range(3):
        print(i*j)
```

Dry-Run

$i=1 \rightarrow 4$
 $j=0 \rightarrow 2$

$i=1 \leftarrow$

$j=0$

$i \times j = 0$

$j=1$

$i \times j = 1$

$j=2$

$i \times j = 1 \times 2 = 2$

$i = 2$

— o/p →

0
1
2
2
2
4

$j = 0$

2

$j = 1$

2

$j = 2$

4

while loop

for loop

Nesting:

①

while

while

②

for

for

③

while

for

④

for

while

while

Nested = 3

```
for i in range(1, 4):
```

```

for i in range(1, 4):
    for j in range(1, 4):
        print(i, j, end=" ")

```

$i=1$

$j=1$

1, 1

$j \neq 2$

1, 1, 1, 2

$j = \cancel{3} \cancel{4}$ → break / stop

1, 1, 1, 2, 1, 3

$i = \cancel{1} \text{ (2)}$

$j=1$

1, 1, 1, 2, 1, 3, 2, 1

$j \neq 2$

1, 1, 1, 2, 1, 3, 2, 1, 2, 2, 2, 3

$$j = 3$$

Patterns

⑦

```

*
x x
x x x
x x x x
x x x x x

```

⑧

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

Jump Statement :-

- 1) Break
- 2) Continue
- 3) Pass ✓